ALGEBRA 2 · UNIT 1: FUNCTIONS & TRANSFORMATIONS

Ride-Share Pricing Lab

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze ZipRide's fares: a piecewise function of miles. They evaluate and graph it, model surge pricing and fees as transformations, compose fare with distance-over-time, find inverses to answer "how far can I go?", compare a competitor, and design a fair pricing plan.

One context covers the function toolkit: piecewise functions, transformations, composition, and inverses. Students see why the order of transformations matters when a fee and a surge are both applied.

Standards Addressed

- F-IF.7b Graph piecewise functions
- F-BF.3 Identify the effect of transformations
- F-BF.1c Compose functions
- F-BF.4 Find inverse functions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a fare screenshot. Part 1 piecewise fare.
Day 2	Part 2 transformations, composition, inverses.
Day 3	Part 3 design and pitch.

Materials

- Student handout
- Graphing calculators or Desmos
- Optional: real ride-share fare screenshots

How the Three Versions Differ

Level	What changes
CP2	Evaluate and graph the piecewise fare, describe a stretch and a shift, and use a composition and an inverse at single values.
CP1	Write the composition and the piecewise inverse as rules, and compare with a competitor algebraically.
Honors	Order of transformations, inverse of the composed function, domain and range restrictions, and a design with continuity.

Watch For (Common Misconceptions)

- Using the first-piece rule for all distances.
- Thinking $1.5(f + 2)$ and $1.5f + 2$ are the same.
- Writing f^{-1} as $1 \div f$.

Extension Ideas

- Compare real fare structures in two cities.
- Add a minimum fare and redraw the graph.
- Model tips as a percent.

Answer Key – CP2

Part 1: The Fare Function

#	Task	Answer
1	Find f(4), f(10), and f(15). Explain which rule you used each time.	f(4) = 2.50 + 4.80 = $7.30 (first rule); f(10) = $14.50 (first rule); f(15) = 14.50 + 4.50 = $19.00 (second rule).
2	Why does the second rule start with 14.50?	A 10-mile ride costs 2.50 + 1.20(10) = $14.50, so the second rule continues from there.
3	Make a table for m = 0, 5, 10, 15, 20 and sketch the graph.	2.50, 8.50, 14.50, 19.00, 23.50; the line gets less steep after 10 miles.
4	What are a reasonable domain and range?	Domain m ≥ 0; range f ≥ 2.50

Part 2: Surges, Fees & Questions

#	Task	Answer
5	During a surge, every fare is multiplied by 1.5: S(m) = 1.5 · f(m). Find S(4). Describe the transformation.	S(4) = 1.5 × 7.30 = $10.95; a vertical stretch by 1.5.
6	ZipRide adds a $2 booking fee: B(m) = f(m) + 2. Describe the transformation and find B(15).	Shift up 2; B(15) = $21.00
7	In traffic, the car travels m(t) = 0.4t miles in t minutes. Find the fare for a 20-minute ride, f(m(20)).	m(20) = 8; f(8) = 2.50 + 9.60 = $12.10
8	A rider has $11.50. Use the inverse idea to find how many miles she can ride (no surge).	2.50 + 1.20m = 11.50 → m = 7.5 miles

Part 3: Design a Competitor

#	Task	Answer
9	Write a two-piece fare function with a different rate after some distance.	Answers vary.
10	Make a table and graph your function with ZipRide's.	Answers vary.
11	For what trip lengths is your app cheaper? Explain how you know.	Answers vary.

Answer Key – CP1

Part 1: The Fare Function

#	Task	Answer
1	Write f(m) as a piecewise function and simplify the second piece to the form am + b.	f(m) = 2.50 + 1.20m (0 ≤ m ≤ 10); f(m) = 0.90m + 5.50 (m > 10)
2	Show that the two pieces meet at m = 10 (the function is continuous).	1.20(10) + 2.50 = 14.50 and 0.90(10) + 5.50 = 14.50
3	Find the average rate of change from m = 5 to m = 15.	(19.00 − 8.50) ÷ 10 = $1.05 per mile
4	Graph f and the surge function 1.5 · f on one set of axes. What changes and what stays the same?	Every y-value is multiplied by 1.5; the break point stays at m = 10 and both pieces get steeper.

Part 2: Compose & Invert

#	Task	Answer
5	With m(t) = 0.4t, write f(m(t)) for the first 25 minutes. What does the slope mean?	f(m(t)) = 2.50 + 0.48t: the fare rises $0.48 per minute.
6	Find $f^{-1}(c)$ for each piece and state the domain of each piece of the inverse.	$f^{-1}(c) = (c - 2.50) \div 1.20$ for $2.50 \le c \le 14.50$; $f^{-1}(c) = (c - 5.50) \div 0.90$ for $c > 14.50$
7	Use f^{-1} to find how far a rider can go for $19.00 and for $11.50.	$19.00 → 15 miles; $11.50 → 7.5 miles
8	A competitor charges g(m) = 3.00 + 1.10m for any distance. For which trip lengths is the competitor cheaper?	Up to 10 mi: 2.50 + 1.20m > 3.00 + 1.10m when m > 5. After 10 mi: 0.90m + 5.50 > 3.00 + 1.10m when m < 12.5. The competitor is cheaper for 5 < m < 12.5.

Part 3: Design a Competitor

#	Task	Answer
9	Write your piecewise function and show it is continuous.	Answers vary.
10	Find exactly where your plan becomes cheaper than ZipRide.	Answers vary.
11	Find the inverse of your function's second piece and use it for a customer question.	Answers vary.

Answer Key – Honors

Part 1: Fare Functions

#	Task	Answer
1	Write f(m) piecewise and find its domain and range. Is f one-to-one? Explain.	Domain $m \geq 0$; range $f \geq 2.50$. Yes, it is always increasing, so it is one-to-one.
2	Compare $F_1(m) = 1.5(f(m) + 2)$ and $F_2(m) = 1.5 \cdot f(m) + 2$. Which is fair to riders, and how much do they differ?	$F_1 - F_2 = 1.5(2) - 2 = \$1.00$ more for every ride with F_1 (the surge also multiplies the fee).
3	Describe F_1 as a sequence of transformations of f, in order.	Shift f up 2, then stretch vertically by 1.5.

Part 2: Composition & Inverse

#	Task	Answer
4	Traffic: $m(t) = 0.4t$ for $0 \leq t \leq 15$ and $m(t) = 6 + 0.8(t - 15)$ for $t > 15$. Write f(m(t)) as a piecewise function of t.	For $t \leq 15$: $m \leq 6$, $f = 2.50 + 0.48t$. For $15 < t \leq 20$: $m = 0.8t - 6 \leq 10$, $f = 2.50 + 1.20(0.8t - 6) = 0.96t - 4.70$. For $t > 20$: $m > 10$, $f = 0.90(0.8t - 6) + 5.50 = 0.72t + 0.10$.
5	Find the inverse of f(m(t)) and use it to find the ride time for a \$19.00 fare.	\$19.00 is in the third piece: $0.72t + 0.10 = 19 \rightarrow t = 26.25$ minutes.
6	Prove that $(f^{-1} \circ f)(m) = m$ for $m > 10$.	$f(m) = 0.90m + 5.50$; $f^{-1}(0.90m + 5.50) = (0.90m + 5.50 - 5.50) \div 0.90 = m$

Part 3: Engineer a Plan

#	Task	Answer
7	Design a two-piece plan that meets every condition. Show your algebra.	Example: $3 + 1.00m$ for $0 \leq m \leq 8$ (11 at 8 mi), then $11 + 0.75(m - 8)$ (20 at 20 mi). Base \$3, continuous.
8	Find where your plan and ZipRide cost the same.	Answers vary.
9	Write the inverse and explain one marketing use.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Pricing plan is a correct piecewise function with a correct inverse or composition used to answer a customer question. Goes beyond the minimum with a creative, realistic choice.	Pricing plan is a correct piecewise function with a correct inverse or composition used to answer a customer question.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Pitch shows the piecewise fare function, its graph, transformations for fees or surges, and a comparison with ZipRide. Polished, organized, and visually engaging.	Pitch shows the piecewise fare function, its graph, transformations for fees or surges, and a comparison with ZipRide. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.